

Module 1.3 – Translation to Linear Constraints

0/1 Decision Variables

For every invigilator i and exam shift j :

$$x_{ij} = \begin{cases} 1, & \text{if invigilator } i \text{ is assigned to shift } j, \\ 0, & \text{otherwise.} \end{cases} \quad x_{ij} \in \{0, 1\}.$$

Clause / Rule Translation

Logical clause / rule	0/1 variables	Linear constraint
$\neg Assign(i, j) \vee \neg Assign(i, k)$	x_{ij}, x_{ik}	$x_{ij} + x_{ik} \leq 1$
$Busy(i, j) \rightarrow \neg Assign(i, j)$	x_{ij}	$x_{ij} = 0$
$\sum_{i \in I} Assign(i, j) = cap(j)$	x_{ij}	$\sum_{i \in I} x_{ij} = cap(j)$
$Assign(i, j) \vee Assign(k, j)$	x_{ij}, x_{kj}	$x_{ij} + x_{kj} \geq 1$
At-most- k : $\sum_j Assign(i, j) \leq k$	x_{ij}	$\sum_j x_{ij} \leq k$

Cardinality / Pseudo-Boolean Constraints

Exact capacity:

$$\sum_{i \in I} x_{ij} = cap(j).$$

At-most- k :

$$\sum_j x_{ij} \leq k.$$

These are cardinality / pseudo-Boolean constraints and are represented directly as linear constraints in the LP formulation.